

Michael C. Lagana

mlagana6@gatech.edu
203 635 3170
linkedin.com/in/mlagana6
mlags101.github.io

EDUCATION

Georgia Institute of Technology | *BS in Aerospace Engineering* **Atlanta, GA** · **Expected May 2027**

- Minor in Robotics, BS/MS Honors Program Cumulative GPA: 3.95
- Related Coursework: Thermodynamics & Fluids, Structural Analysis, Jet & Rocket Propulsion

EXPERIENCE

AV, MacCready Works | *Systems Test Engineering Intern* **Moorpark, CA** · **Jun – Aug 2026**

- Prototyped sUAS airframes through rapid CAD iteration and in-house fabrication, from concept to flight-ready hardware
- Supported flight test operations, including test setup and hardware turnaround between flights
- Developed modal testing procedures for the Skyfall Mars helicopter program
- Adapted a motion capture drop test rig for a new vehicle and documented data collection and processing procedures

Aerospace Robotics Laboratory | *Undergraduate Researcher* **Atlanta, GA** · **Aug 2025 – Present**

- Designed and fabricated autonomy-capable quadcopter testbeds for research in swarm coordination, mapping, and localization
- Developed flight code and procedures integrating quadcopters with a motion capture system for high-precision localization
- Developing a small underwater vehicle to enable research in autonomous underwater manipulation

GTRI, Advanced Combat Lab | *Electrical Engineering Co-Op* **Atlanta, GA** · **Jan – Aug 2025**

- Diagnosed and repaired UAV avionics faults through field RF testing and signal analysis
- Designed NMEA and Ethernet converter PCBs in Altium for deployment in marine environments
- Built an engine test rig and collected data to predict failure modes, authoring test procedures and reports
- Modeled engine cut-off failure in MATLAB, producing estimates that informed safe design decisions
- Designed assembly jigs in SolidWorks to ensure safe, repeatable custom alternator assembly

RoboNav | *Drone Lead Engineer* **Atlanta, GA** · **Oct 2023 – Present**

- Founded the drone team, secured sponsorship, and directed 10+ members through development of EEVi v3
- Designed a 2-axis auto-tracking antenna in SolidWorks running Ardupilot, extending comms range by 2 km
- Developed a drone frame from scrapped carbon fiber with parametric SolidWorks components for desert operation
- Evaluated landing gear structural viability through FEA in ANSYS Mechanical

SKILLS

Languages: Python, MATLAB, C++

CAD & Simulation: SolidWorks, ANSYS FEA & CFD, Altium, OpenVSP, Simulink

Robotics & Embedded: ROS2, Ardupilot, STM32CubeMX, Arduino, Linux, Git/GitHub, Motion Capture

Fabrication: Lathe, Mill, CNC, Waterjet, MIG Welding, PCB Fabrication, FDM/SLA/SLS